ESE382 Digital System Design Using VHDL and PLDs

Spring 26, Ken Short
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Laboratory 11: Signature Analyzer II - Linear Feedback Shift Register and Cyclic Redundancy Check (CRC)

To be performed the week starting April 26th.

Prerequisite Reading
1. Chapter 9 of the textbook.
2. HP 5004A Signature Analyzer Operating and Service Manual

Purpose
The HP 5004A **Signature Analyzer** is a 1970s-era digital troubleshooting tool used to diagnose complex logic circuits down to the component level. It compresses long bit streams at a digital circuit node into a unique 4-digit hexadecimal "signature" (using a 16-bit cyclic redundancy check (CRC)). This signature allows a technician to compare faulty circuit nodes against a list of know good signatures to determine which ICs on a circuit board are faulty. Signature analyzers are primarily used for servicing digital equipment (e.g., microprocessors, arcade boards) where conventional scopes or voltmeters are ineffective.

An HP signature analyzer works by capturing data streams triggered by start, stop, and clock signals, then displaying a 4 hex-character signature on a seven-segment display using the funny he font.

The signature (the 16-CRC) can be computed in hardware using a 16-bit **linear feedback shift register (LFSR)**. An LFSR is a hardware or software shift register whose input bit is driven by a linear function—usually the XOR of specific bits, known as "taps." It generates a long, seemingly random sequence of bits (**pseudorandom**) very quickly, making it ideal for cryptography, digital signature analysis, and pseudo-noise sequence generation. When used as a signature analyzer the LFSR includes the data stream XORed with feedback input.

HP selected the polynomial $x^{16}+x^{12}+x^9+x^7+1$ instead of a standard CRC-16 or CCITT-16 polynomial, in order to scatter the missed errors as much as possible and to avoid selecting feedback taps that are evenly spaced or four or eight bits apart because the types of instruments that would be most frequently tested tend to repeat patterns at four and eight-bit intervals.

This 16-bit CRC is claimed to have a 99.998% probability of detecting even a single-bit error in a long data stream.

Tasks
Task 1: hp5004a_lfsr
Task 2: hp5004a_SA_w_display

Design Tasks
You must design the two design entities.

Design Task 1: hp5004a_lfsr
This design entity is a 0-based implementation of a CRC system based on the polynomial $x^{16}+x^{12}+x^9+x^7+1$ that HP uses in its HP 5004A signature analyzer. The architecture must be behavioral. `rst_bar` must be synchronous. The entity declaration is:

```
entity hp5004a_lfsr is
    Port (
        clk      : in  std_logic;  -- Logic clock from the circuit under test
        rst_bar  : in  std_logic;  -- Reset or "Start" pulse
        gate     : in  std_logic;  -- High when sampling data (the measurement window)
        data_in  : in  std_logic;  -- Data probe input
        sig_out  : out std_logic_vector(15 downto 0) -- 16-bit raw signature
        );
end hp5004a_lfsr;
```

Write a design description and non-self-checking testbench for this entity. Use your testbench to verify your design.

Submit your properly commented design description and testbench source files as part of your prelab.

Design Task 2: hp5004a_SA_w_display
The ***multiplexed_display_driver*** from Laboratory 10 is to be combined with the ***hp5004a_lfsr*** from Task 1 of this laboratory to create a system that computes and displays the signature of a sampled data stream. You must determine from the two entities being combined what the entity declaration for the top-level entity must be.

Write a structural design description and non-self-checking testbench for the top-level entity. Use your testbench to verify your design.

Laboratory Tasks

Laboratory Task 1: hp5004a_lfsr
Import your VHDL design description and testbench files for this design entity. Use your **test-bench** to verify your design.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.

Laboratory Task 2: hp5004a_SA_w_display
Import your VHDL design description files for this design. Use your **testbench** to verify your design.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.

Synthesize and place and route your `hp5004a_SA_w_display` design description for an ispMACH4A5-64/32-10JC CPLD target.

Demonstrate your timing simulation to a TA and have the TA verify and sign that your timing simulation was successful.

Program your design into an ispMACH4A5-64/32-10JC CPLD.

Have the TA verify and sign that your programmed CPLD operates correctly in the test fixture provided.